

ECS 409/609

Assignment 1

Sequential Construct-I

Integer Arithmetic and Logical Instructions

Kunwar Arpit Singh

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For code repository of this assignment:  **MIPS Assignments**

1. Problem 1

For a given arithmetic and geometric progression sequence:

- 1, 11, 21, 31, ... (A.P.)
- 4, 8, 16, ... (G.P.)

Write a MIPS assembly program to compute the eighth term (in A.P.) / fourth term (in G.P.) and sum of the first six terms (in A.P.) / first four numbers (in G.P.). The initial values are stored in registers or memory.

Part A: Arithmetic Progression

MIPS Assembly Source

```
1 # Assignment 1: Sequential Construct-I
2 # Programmed By : Kunwar Arpit Singh
3
4 .text
5 .globl main
6
7 main:
8     # $t0 = a
9     li $t0, 1
10
11     # $t1 = d
12     li $t1, 10
13
14     # Calculate the 8th term
15     # Formula:  $a_n = a + (n-1)d$ 
16     li $t2, 8 # n = 8
17     addi $t3, $t2, -1 # Calculate (n-1)
18     mul $t4, $t3, $t1 # Calculate (n-1) * d
19     add $s0, $t0, $t4 # Result:  $a + (n-1)d$ . Stored in $s0
20
21     # Calculate the sum of the first 6 terms
22     # Formula:  $S_n = (n/2) * (2a + (n-1)d)$ 
```

```

23      li $t2, 6 # n = 6
24
25      # Calculate (n-1)d part
26      addi $t3, $t2, -1 # Calculate (n-1)
27      mul $t4, $t3, $t1 # Calculate (n-1) * d
28
29      # Calculate 2a part
30      li $t8, 2 # Load immediate value 2 into a register
31      mul $t5, $t0, $t8 # Correctly use mul with registers: 2 * a
32
33      # Calculate (2a + (n-1)d)
34      add $t6, $t5, $t4
35
36      # Calculate n/2 part
37      div $t2, $t8 # n / 2. Quotient is stored in special register 'lo'
38      mflo $t7 # Move the quotient (n/2) from 'lo' into $t7
39
40      # Final sum calculation
41      mul $s1, $t7, $t6 # (n/2) * (2a + (n-1)d). Result stored in $s1
42
43      li $v0, 10
44      syscall

```

Output Screenshots

Register values (A.P.)

Int Regs [10]	
PC	= 4194404
EPC	= 0
Cause	= 0
BadVAddr	= 0
Status	= 805371664
HI	= 0
LO	= 156
R0 [r0]	= 0
R1 [at]	= 0
R2 [v0]	= 10
R3 [v1]	= 0
R4 [a0]	= 4
R5 [a1]	= 2147479308
R6 [a2]	= 2147479328
R7 [a3]	= 0
R8 [t0]	= 1
R9 [t1]	= 10
R10 [t2]	= 6
R11 [t3]	= 5
R12 [t4]	= 50
R13 [t5]	= 2
R14 [t6]	= 52
R15 [t7]	= 3
R16 [s0]	= 71
R17 [s1]	= 156
R18 [s2]	= 0
R19 [s3]	= 0
R20 [s4]	= 0
R21 [s5]	= 0
R22 [s6]	= 0
R23 [s7]	= 0
R24 [t8]	= 2
R25 [t9]	= 0
R26 [k0]	= 0
R27 [k1]	= 0
R28 [gp]	= 268468224
R29 [sp]	= 2147479304
R30 [s8]	= 0
R31 [ra]	= 4194328

Part B: Geometric Progression

MIPS Assembly Source

```
1 # Assignment 1: Sequential Construct-I
2 # Programmed By : Kunwar Arpit Singh
3
4 .text
5 .globl main
6
7 main:
8     # $t0 = a=
9     li $t0, 4
10
11     # $t1 = r=
12     li $t1, 2
13
14     # Calculate the 4th term
15     # Formula:  $a_n = a * r^{(n-1)}$ 
16     # For n=4, this is  $a * r^3$ 
17     mul $t3, $t1, $t1 # $t3 =  $r^2$ 
18     mul $t3, $t3, $t1 # $t3 =  $r^3$ 
19     mul $s2, $t0, $t3 # Result:  $a * r^3$ . Stored in $s2
20
21     # Calculate the sum of the first 4 terms
22     # Formula:  $S_n = a * (r^n - 1) / (r - 1)$ 
23     # For n=4, this is  $a * (r^4 - 1) / (r - 1)$ 
24
25     # Calculate  $r^n - 1$  part
26     mul $t4, $t3, $t1 # $t4 =  $r^4$  (since $t3 holds  $r^3$ )
27     addi $t5, $t4, -1 # $t5 =  $r^4 - 1$ 
28
29     # Numerator:  $a * (r^n - 1)$ 
30     mul $t6, $t0, $t5
31
32     # Denominator:  $r - 1$ 
33     addi $t7, $t1, -1
34
35     # Final division
36     div $t6, $t7 # Numerator / Denominator. Quotient stored in 'lo'
37     mflo $s3 # Move result from 'lo' into $s3
38
39     li $v0, 10
40     syscall
```

Output Screenshots

Register values (G.P.)

Int Regs [10]		
PC	=	4194388
EPC	=	0
Cause	=	0
BadVAddr	=	0
Status	=	805371664
HI	=	0
LO	=	60
R0 [r0]	=	0
R1 [at]	=	0
R2 [v0]	=	10
R3 [v1]	=	0
R4 [a0]	=	4
R5 [a1]	=	2147479308
R6 [a2]	=	2147479328
R7 [a3]	=	0
R8 [t0]	=	4
R9 [t1]	=	2
R10 [t2]	=	0
R11 [t3]	=	8
R12 [t4]	=	16
R13 [t5]	=	15
R14 [t6]	=	60
R15 [t7]	=	1
R16 [s0]	=	0
R17 [s1]	=	0
R18 [s2]	=	32
R19 [s3]	=	60
R20 [s4]	=	0
R21 [s5]	=	0
R22 [s6]	=	0
R23 [s7]	=	0
R24 [t8]	=	0
R25 [t9]	=	0
R26 [k0]	=	0
R27 [k1]	=	0
R28 [gp]	=	268468224
R29 [sp]	=	2147479304
R30 [s8]	=	0
R31 [ra]	=	4194328

2. Problem 2

Write a MIPS assembly program to implement a half adder and a half subtractor using logical instructions.

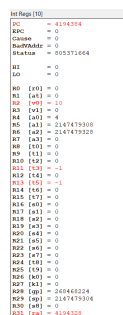
$A = 0 ; B = 0$

MIPS Assembly Source

```
1 # Assignment 1: Sequential Construct-I
2 # Programmed By : Kunwar Arpit Singh
3
4 .text
5 .globl main
6
7 main:
8     # Case 1: A = 0, B = 0
9     li $t0, 0 # A = 0
10    li $t1, 0 # B = 0
11
12    # Half Adder Logic (Sum = A XOR B, Carry = A AND B)
13    nor $t3, $t1, $t1 # NOT B
14    and $t4, $t0, $t3 # A AND (NOT B)
15    nor $t5, $t0, $t0 # NOT A
16    and $t6, $t5, $t1 # (NOT A) AND B
17    or $s0, $t4, $t6 # Sum
18    and $s1, $t0, $t1 # Carry
19
20    # Half Subtractor Logic (Difference = A XOR B, Borrow = (NOT A) AND B)
21    move $s2, $s0 # Difference
22    move $s3, $t6 # Borrow
23
24    li $v0, 10
25    syscall
```

Output Screenshots

Register values (Half Adder/Subtractor)



The screenshot displays the MIPS register file with the following values:

Register	Value
\$0	0
\$1	0
\$2	0
\$3	0
\$4	0
\$5	0
\$6	0
\$7	0
\$8	0
\$9	0
\$10	0
\$11	0
\$12	0
\$13	0
\$14	0
\$15	0
\$16	0
\$17	0
\$18	0
\$19	0
\$20	0
\$21	0
\$22	0
\$23	0
\$24	0
\$25	0
\$26	0
\$27	0
\$28	0
\$29	0
\$30	0
\$31	0

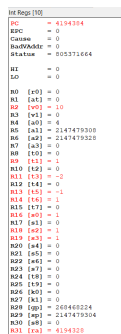
A = 0 ; B = 1

MIPS Assembly Source

```
1 # Assignment 1: Sequential Construct-I
2 # Programmed By : Kunwar Arpit Singh
3
4 .text
5 .globl main
6
7 main:
8     # Case 1: A = 0, B = 0
9     li $t0, 0 # A = 0
10    li $t1, 1 # B = 1
11
12    # Half Adder Logic (Sum = A XOR B, Carry = A AND B)
13    nor $t3, $t1, $t1 # NOT B
14    and $t4, $t0, $t3 # A AND (NOT B)
15    nor $t5, $t0, $t0 # NOT A
16    and $t6, $t5, $t1 # (NOT A) AND B
17    or $s0, $t4, $t6 # Sum
18    and $s1, $t0, $t1 # Carry
19
20    # Half Subtractor Logic (Difference = A XOR B, Borrow = (NOT A) AND B)
21    move $s2, $s0 # Difference
22    move $s3, $t6 # Borrow
23
24    li $v0, 10
25    syscall
```

Output Screenshots

Register values (Half Adder/Subtractor)



The screenshot shows the MIPS register file with the following values:

Register	Value
\$0	0
\$1	1
\$2	0
\$3	0
\$4	0
\$5	0
\$6	0
\$7	0
\$8	0
\$9	0
\$10	0
\$11	0
\$12	0
\$13	0
\$14	0
\$15	0
\$16	0
\$17	0
\$18	0
\$19	0
\$20	0
\$21	0
\$22	0
\$23	0
\$24	0
\$25	0
\$26	0
\$27	0
\$28	0
\$29	0
\$30	0
\$31	0

A = 1 ; B = 0

MIPS Assembly Source

```

1 # Assignment 1: Sequential Construct-I
2 # Programmed By : Kunwar Arpit Singh
3
4 .text
5 .globl main
6
7 main:
8     # Case 1: A = 0, B = 0
9     li $t0, 1 # A = 1
10    li $t1, 0 # B = 0
11
12    # Half Adder Logic (Sum = A XOR B, Carry = A AND B)
13    nor $t3, $t1, $t1 # NOT B
14    and $t4, $t0, $t3 # A AND (NOT B)
15    nor $t5, $t0, $t0 # NOT A
16    and $t6, $t5, $t1 # (NOT A) AND B
17    or $s0, $t4, $t6 # Sum
18    and $s1, $t0, $t1 # Carry
19
20    # Half Subtractor Logic (Difference = A XOR B, Borrow = (NOT A) AND B)
21    move $s2, $s0 # Difference
22    move $s3, $t6 # Borrow
23
24    li $v0, 10
25    syscall

```

Output Screenshots

Register values (Half Adder/Subtractor)

```

int Regs [10]
PC = 4194304
EPC = 0
Cause = 0
BadVAddr = 0
Status = 805371664
MT = 0
MD = 0
NO [x0] = 0
R1 [a1] = 0
R2 [v0] = 20
R3 [v1] = 0
R4 [a0] = 4
R5 [a1] = 2147479308
R6 [a2] = 2147479328
R7 [a3] = 0
R8 [t0] = 1
R9 [t1] = 0
R10 [t2] = 0
R11 [t3] = -1
R12 [t4] = -1
R13 [t5] = -2
R14 [t6] = 0
R15 [t7] = 0
R16 [t8] = 1
R17 [t9] = 0
R18 [t10] = 1
R19 [t11] = 0
R20 [t12] = 0
R21 [t13] = 0
R22 [t14] = 0
R23 [t15] = 0
R24 [t16] = 0
R25 [t17] = 0
R26 [t18] = 0
R27 [t19] = 0
R28 [t20] = 268468224
R29 [t21] = 2147479304
R30 [t22] = 0
R31 [ra] = 4194328

```

A = 1 ; B = 1

MIPS Assembly Source

```

1 # Assignment 1: Sequential Construct-I
2 # Programmed By : Kunwar Arpit Singh
3

```


3. Problem 3

Problem statement

Write a MIPS assembly program to swap the values of two registers using only logical instructions.

MIPS Assembly Source

```
1  # Assignment 1: Sequential Construct-I
2  # Programmed By : Kunwar Arpit Singh
3
4  .text
5  .globl main
6
7  main:
8      li $s0, 12345
9      li $s1, 56789
10
11     # s0 = s0 XOR s1
12     and $t0, $s0, $s1
13     nor $t1, $t0, $t0
14     or $t2, $s0, $s1
15     and $s0, $t2, $t1
16
17     # s1 = s1 XOR s0
18     and $t0, $s1, $s0
19     nor $t1, $t0, $t0
20     or $t2, $s1, $s0
21     and $s1, $t2, $t1
22
23     # s0 = s0 XOR s1
24     and $t0, $s0, $s1
25     nor $t1, $t0, $t0
26     or $t2, $s0, $s1
27     and $s0, $t2, $t1
28
29     li $v0, 10
30     syscall
```

Output Screenshots

Register values (Register Swap)

Int Regs [10]		
PC	=	4194400
EPC	=	0
Cause	=	0
BadVAddr	=	0
Status	=	805371664
HI	=	0
LO	=	0
R0	[r0]	= 0
R1	[at]	= 0
R2	[v0]	= 10
R3	[v1]	= 0
R4	[a0]	= 4
R5	[a1]	= 2147479316
R6	[a2]	= 2147479336
R7	[a3]	= 0
R8	[t0]	= 8232
R9	[t1]	= -8233
R10	[t2]	= 65021
R11	[t3]	= 0
R12	[t4]	= 0
R13	[t5]	= 0
R14	[t6]	= 0
R15	[t7]	= 0
R16	[s0]	= 56789
R17	[s1]	= 12345
R18	[s2]	= 0
R19	[s3]	= 0
R20	[s4]	= 0
R21	[s5]	= 0
R22	[s6]	= 0
R23	[s7]	= 0
R24	[t8]	= 0
R25	[t9]	= 0
R26	[k0]	= 0
R27	[k1]	= 0
R28	[gp]	= 268468224
R29	[sp]	= 2147479312
R30	[s8]	= 0
R31	[ra]	= 4194328

4. Problem 4

Problem statement

Write a MIPS assembly program to compute a weighted average of four 32-bit numbers (A, B, C, and D) using the formula: $(0.125A + 0.25B + 0.5C + 0.5D)$. Do not use multiplication, division, or floating-point instructions. The final value should be returned in register \$a0.

MIPS Assembly Source

```
1  # Assignment 1: Sequential Construct-I
2  # Programmed By : Kunwar Arpit Singh
3
4  .text
5  .globl main
6
7  main:
8      li $t0, 800
9      li $t1, 400
10     li $t2, 200
11     li $t3, 100
12
13     # Calculate the weighted terms.
14     # The formula is  $(0.125A + 0.25B + 0.5C + 0.5D)$ .
15     # This is achieved using bit shifts, which is allowed and avoids the
        forbidden 'mult'/'div' instructions.
16
17     srl $t4, $t0, 3 # $t4 = A / 8
18     srl $t5, $t1, 2 # $t5 = B / 4
19     srl $t6, $t2, 1 # $t6 = C / 2
20     srl $t7, $t3, 1 # $t7 = D / 2
21
22     # Sum the individual terms to get the final weighted average.
23     add $s0, $t4, $t5
24     add $s0, $s0, $t6
25     add $s0, $s0, $t7
26
27     # Move the final result into the specified return register, $a0.
28     move $a0, $s0
29
30     li $v0, 10
31     syscall
```

Output Screenshots

Register values (Weighted Average)

Int Regs [10]		
PC	=	4194392
EPC	=	0
Cause	=	0
BadVAddr	=	0
Status	=	805371664
HI	=	0
LO	=	0
R0 [r0]	=	0
R1 [at]	=	0
R2 [v0]	=	10
R3 [v1]	=	0
R4 [a0]	=	350
R5 [a1]	=	2147479316
R6 [a2]	=	2147479336
R7 [a3]	=	0
R8 [t0]	=	800
R9 [t1]	=	400
R10 [t2]	=	200
R11 [t3]	=	100
R12 [t4]	=	100
R13 [t5]	=	100
R14 [t6]	=	100
R15 [t7]	=	50
R16 [s0]	=	350
R17 [s1]	=	0
R18 [s2]	=	0
R19 [s3]	=	0
R20 [s4]	=	0
R21 [s5]	=	0
R22 [s6]	=	0
R23 [s7]	=	0
R24 [t8]	=	0
R25 [t9]	=	0
R26 [k0]	=	0
R27 [k1]	=	0
R28 [gp]	=	268468224
R29 [sp]	=	2147479312
R30 [s8]	=	0
R31 [ra]	=	4194328